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**The Role of CSD2 in RyR3-Mediated Calcium
Release in Wolfram Syndrome Type 2**

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1. Abstract

This project investigates the physical interaction between CSD2 and type 3 ryanodine receptors (RyR3), as well as the effect of CSD2 on RyR3-mediated calcium release. The study aims to clarify a possible mechanism of Ca^{2+} dysregulation in Wolfram syndrome type 2 (WS2), arising from the influence of CSD2 on RyR3. WS2 is caused by a loss of function mutation in the *CSD2* gene. CSD2 is a multifunctional protein involved in the regulation of redox reactions and reactive oxygen species, autophagy, and calcium homeostasis and signalling. Dysregulation of these processes leads to clinical features such as diabetes mellitus, optic atrophy, and deafness, symptoms shared with WS type 1, as well as platelet aggregation defects and gastrointestinal ulcers, which are specific to WS2.

The physical interaction between CSD2 and RyR3 was demonstrated through Co-Immunoprecipitation assays using anti-RyR3 antibodies and confirmed by immunoblotting. The role of CSD2 in RyR3-mediated Ca^{2+} flux was assessed by measuring caffeine induced ER Ca^{2+} release in CSD2 knockout cells using fluorescence microscopy with FURA2-AM. ER Ca^{2+} storage capacity was also evaluated in the presence or absence of CSD2 through inhibition of SERCA with thapsigargin. In addition, the impact of two CSD2 mutants (CSD2^{FeSCD} and CSD2^{N72S}) on RyR3 function was examined by transfecting CSD2 knockout cells with mCherry vectors carrying the mutant forms, allowing for selection of transfected cells during microscopy experiments. Finally, the consequences of CSD2 loss on mitochondrial calcium dynamics were investigated in CSD2 knockout cells using the mtCEPIA fluorophore and caffeine as an RyR3 agonist.

In conclusion, the results demonstrate that CSD2 physically and functionally interacts with RyR3, acting in its wild type form as a negative regulator without affecting the overall Ca^{2+} content of the ER. Moreover, CSD2 is essential for maintaining calcium uptake into mitochondria. In contrast, CSD2 mutants do not fully reproduce the effects of CSD2 loss, suggesting that they may influence additional pathways that warrant further investigation.

2. Introduction

Wolfram syndrome (WS) is a rare neurodegenerative disorder transmitted in an autosomal recessive way. It was first described in 1938 by Wolfram and Wagener, who reported the co-occurrence of diabetes mellitus and optic atrophy in four siblings of consanguineous parents[1]. Typically, during the course of the disease, these initial two symptoms are followed by other specific clinical features that together can be summarised with the acronym DIDMOAD: diabetes insipidus (DI), diabetes mellitus (DM), optic atrophy (OA), and deafness (D)[2].

However, WS is frequently complicated by other symptoms which can differ on the specific type of WS: Wolfram syndrome type 1 (WS1) and Wolfram Syndrome type 2 (WS2). Psychiatric, endocrinological, and neurological disorders are more common in WS1 [Table 1], while urinary tract abnormalities, gastrointestinal ulcers and impaired platelet aggregation are particularly described in WS2 [Table 2][3].

Nowadays, the diagnosis of WS is guided by the observation of hallmark symptoms DM and OA, and is confirmed through genetic screening for loss-of-function mutations in the *WFS1* and *CISD2* genes, resulting in WS1 and WS2, respectively[4]. Prognosis for both WS1 and WS2 is always poor, and it heralds a premature death at a mean age of 30 years (25-49 years).

Major Clinical Features	Age at Diagnosis	Percentage of WS1 Patients with a Single Disorder
Diabetes insipidus	14 years (3 months–40 years)	37.7%
Diabetes mellitus	6 years (3 weeks–16 years)	98.2%
Optic atrophy	11 years (6 weeks–19 years)	82.1%
Sensorineural hearing loss	12.5 years (5–39 years)	48.2%
Neurological and autonomic disorders (central apnea, ataxia, dysphagia, areflexia, epilepsy, decreased ability to taste and detect odors, headaches, orthostatic hypotension, hyperpyrexia, hypothermia, constipation, gastroparesis)	16 years (5–44 years)	17.1%
Urinary tract complications (neurogenic bladder, bladder incontinence, urinary tract infections)	20 years (13–33 years)	19.4%
Other common clinical features		
Psychiatric symptoms (anxiety, panic attacks, depression, mood swings, sleep abnormalities, psychosis)	20.5 years (17–23 years)	44.4%
Endocrinological disorders (hypogonadism, deficient growth hormone secretion, corticotropin deficiency, delayed menarche in female)	8 years (7–9 years)	6.6%

Table 1. Percentage and age of diagnosis of WS1 major clinical features, Rigoli et al., *Int. J. Environ. Res. Public Health*, 2022.

2.1 Wolfram syndrome type 1 (WS1)

WS1 is the most common type of WS with the highest prevalence observed in Lebanon (1:68,000) and in a small area of Sicily, in Italy, (1:54,478), geographical regions with high rates of consanguinity in marriage[5, 6].

Mutations in the WFS1 gene, located on chromosome 4p16, are responsible for this disease. This gene encodes wolframin (WFS1), a transmembrane glycoprotein localized in the mitochondria-associated endoplasmic reticulum (ER) membranes (MAMs)[7, 8].

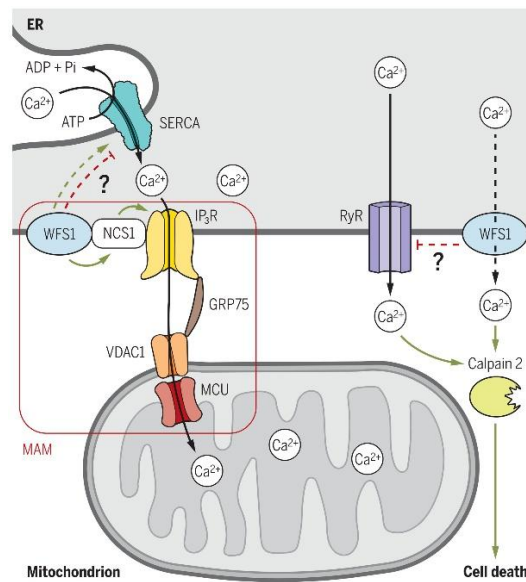


Figure 1. Schematic overview of possible effects of WFS1 on IP₃R, SERCA and RyR. MAMs are indicated by red square. Loncke et al., *Sci. Signal*, 2021.

WFS1 plays a key role in maintaining ER Ca^{2+} homeostasis. Although it reduces SERCA abundance, likely via proteasomal degradation, WFS1 is nevertheless required to sustain proper ER Ca^{2+} loading, as its deficiency leads to impaired SERCA function and pathological ER Ca^{2+} leakage[9]. Further studies have also implicated ryanodine receptors (RyRs) in this leak, which will be discussed in detail later.

Furthermore, WFS1 appears to modulate IP₃R activity. Specifically, it may enhance IP₃ sensitivity by forming a molecular complex with IP₃R and the neuronal Ca^{2+} sensor 1 (NCS1), thereby increasing IP₃R-mediated Ca^{2+} release.[10]. Conversely, loss of WFS1 function reduces IP₃R activity, impairs ER–

mitochondrial interactions, and decreases Ca^{2+} transfer to mitochondria ending in compromised mitochondrial bioenergetics and function [Fig. 1][7].

In addition, mutations in WFS1 can lead to an accumulation of misfolded proteins and the subsequent activation of the unfolded protein response (UPR) to restore ER homeostasis[11].

However, prolonged activation of the UPR, as occurs in chronic ER stress conditions, ultimately triggers cell apoptosis[12, 13]. In this manner, loss of WFS1 function leads to cell death.

2.2 Wolfram syndrome type 2 (WS2)

The second type of WS is caused by loss of function mutations in the CDGSH iron-sulfur domain-containing protein 2 gene (*CISD2* gene)[14].

Unfortunately, because of the extremely low number of reported cases, it is not possible to establish the epidemiology of this condition[3]. Nevertheless, data from patients with a confirmed molecular diagnosis of WS2 have highlighted two clinical features that distinguish it from WS1[3].

The first regards gastrointestinal disorders, manifesting as recurrent upper gastrointestinal bleeding ulcers[3]. This condition is often discovered later than the glucose metabolism impairment, present also in WS1[2].

Gastrointestinal bleedings often relate to a coagulation cascade problem which is the second distinctive feature of WS2 [Table 2]. About 85% of the patients show prolonged bleeding due to a platelet aggregation deficit[3, 15].

Clinical Features	WS1	WS2
Diabetes mellitus	Yes	Yes
Optic atrophy/neuropathy	Yes	Yes
Sensorineural deafness	Yes	Yes
Diabetes insipidus	Yes	No (sporadic cases described)
Other endocrinological abnormalities	Yes	No (sporadic cases described)
Neurological disorders	Yes	No (sporadic cases described)
Psychiatric symptoms	Yes	No (mood disorders described)
Genitourinary problems	Yes	Yes
Platelet aggregation defects	No	Yes
GI ulcers	No	Yes

Table 2. Comparison of different clinical features in WS1 and WS2 F. M. Rosanio et al., *Int. J. Environ. Res. Public Health*, 2022.

2.3 CISD2

The CDGSH iron-sulfur domain-containing protein 2 gene (CISD2 gene) is located on chromosome 4q24 and consists of 3 exons encoding the homodimer protein CISD2 also known under the aliases of ERIS, Naf-1, Miner1, Noxp70 or WSF2[3, 16].



Figure 2. CISD2 structure from Protein Data Bank (www.rcsb.org), PDB ID: 4007.

CISD2 is a dimeric transmembrane protein and, as indicated by its name, it contains a cytosolic CSDGH domain which binds to a redox-active 2Fe-2S cluster domain at the C terminus [17].

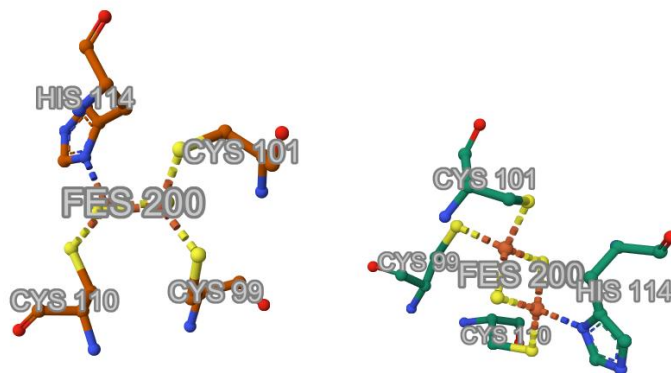


Figure 3. CISD2 2Fe-2S cluster domain structure from Protein Data Bank (www.rcsb.org), PDB ID: 4007.

The N-terminal portion of CISD2, which includes the transmembrane region, is essential for its targeting to the endoplasmic reticulum and the MAMs and for mediating interactions with Bcl-2 and other Ca^{2+} regulators[16, 18].

Additionally, it contains a conserved amino acid sequence that classifies it within the NEET protein family, which are involved in iron management and the generation of reactive oxygen species (ROS)[3, 19].

CISD2 is a protein with a ubiquitous expression underlying its role in maintaining cellular homeostasis in various cell types and organs[20]. Like WSF1, it is located on the ER membrane at the level of MAMs, which has prompted investigations into potentially similar roles in cellular homeostasis[18]. On the other hand, while WSF1 has been exhaustively studied, several aspects of CISD2 are not fully understood[21].

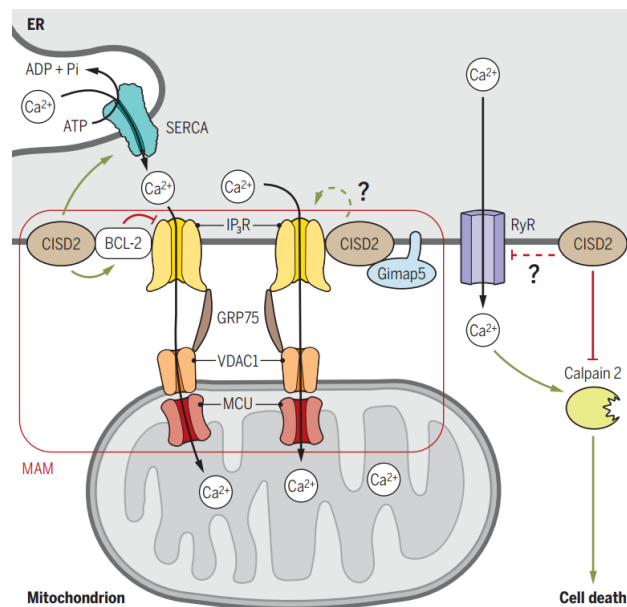


Figure 4. MAMs complex and possible interactions of CISD2 with IP₃R, SERCA and RyR Loncke et al., *Sci. Signal*, 2021.

Multiple functions of CISD2 have been reported. Owing to its 2Fe-2S cluster domain, it acts as a regulator of redox reactions and reactive oxygen species (ROS) management[22, 23]. Indeed, it has been shown that CISD2 deficiency increases cell death via ferroptosis, due to elevated iron levels in mitochondria[3].

Conversely, CISD2 overexpression appears to exert a neuroprotective effect in Alzheimer's disease mouse models by preventing mitochondrial damage, while in cardiac cells it delays cellular aging[24, 25]. Furthermore, supporting its role in cell growth and proliferation, CISD2 has been abundantly found in certain cancer cells[26].

In addition, C1SD2 impacts autophagy and Ca^{2+} homeostasis through its interaction with Bcl-2. This interaction is required to inhibit BECLIN 1-mediated autophagy, but it also indirectly modulates the effect of Bcl-2 on IP_3R Ca^{2+} release in the cytosol and preserves ER-mitochondrial contact sites[27].

Moreover, C1SD2 regulates Ca^{2+} signalling and homeostasis by acting as a regulator of SERCA, thereby ensuring proper ER Ca^{2+} filling[25, 28]. Currently, no evidence suggests a direct modulation of RyRs by C1SD2.

In WS2 patients, potentially all these functions of C1SD2 are dysregulated or lost. To date, four mutations have been associated with WS2 disease. The first two result in aberrantly spliced mRNA due to skipping of exon 1 or exon 2, respectively[29, 30]. The third mutation, a c.215 A>G point mutation (p.Asn72Ser) in exon 2, produces the C1SD2^{N72S} variant, which lacks the 2Fe-2S cluster domain[31]; this variant will be used in the present study. The fourth mutation results in the deletion of exon 2 and likely of an unknown part of exon 3[32]. A fifth identified mutation is a frameshift mutation in exon 2.[33].

2.4 Calcium signalling and homeostasis

Ionized calcium (Ca^{2+}) is one of the most versatile intracellular messengers. It plays a fundamental role in maintaining cellular homeostasis, mediating responses to external stimuli, and regulating physiological processes both in the cytosol and the nucleus[34].

The resting or basal cytosolic concentration of Ca^{2+} is approximately 100 nM. Upon stimulation, this concentration can rapidly increase to 0.5–1 μM . The rise in cytosolic Ca^{2+} originates from two main sources: release from intracellular stores or influx from the extracellular space. Although both mechanisms are employed, the relative contribution varies depending on the cell type[34].

Among the principal Ca^{2+} - releasing channels located on intracellular stores are the inositol 1,4,5-trisphosphate receptors (IP_3Rs)[35]. These receptors are activated by inositol 1,4,5-trisphosphate (IP_3), generated after the stimulation of cell surface receptors (such as G-protein–coupled receptors (GPCR) or receptor

tyrosine kinases) that activate PLC- β , which hydrolyses phosphatidylinositol 4,5-bisphosphate (PIP₂) into diacylglycerol (DAG) and IP₃[34, 35].

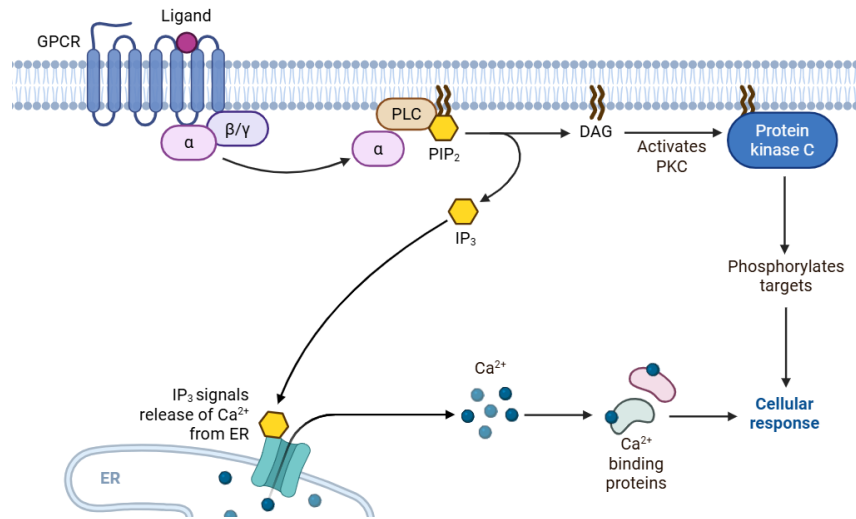


Figure 5. schematic representation IP₃ – mediated Ca²⁺ release. Ligand activated GPCR dissociates Gα subunit activating PLC. PLC hydrolyses PIP₂ in IP₃ and DAG. IP₃ directly activates IP₃R Ca²⁺ release while DAG activates PKC phosphorylation pathways. The figure was retrieved from BioRender.com.

In this manner, IP₃R releases an IP₃ – mediated Ca²⁺ flux from the Golgi, nuclear envelope and ER[35]. The latter is the main cell Ca²⁺ reservoir with an inner Ca²⁺ concentration 1000x higher than in the cytosol, maintained by the active transport of Ca²⁺ of the sarco/endoplasmic Ca²⁺ ATPases (SERCAs)[36, 37].

On the ER membrane, IP₃Rs are closely associated with other regulators of Ca²⁺ dynamics, including mitochondria-associated membranes (MAMs)[38, 39]. MAMs are crucial sites of communication between the ER and mitochondria, allowing direct transfer of Ca²⁺ and thereby supporting mitochondrial function[40, 41]. Mitochondrial Ca²⁺ uptake is essential for ATP production and cell survival, while insufficient Ca²⁺ delivery promotes autophagy[42, 43]. Conversely, excessive mitochondrial Ca²⁺ accumulation can increase the permeability of the inner mitochondrial membrane (IMM), leading to cell death[44]. Thus, ER–mitochondrial Ca²⁺ fluxes have a key role in cell fate [39].

At the MAM interface, the ER and the outer mitochondrial membrane (OMM) are separated by a distance of 10–80 nm[41]. In this region, IP₃Rs are physically linked to voltage-dependent anion channels (VDACs) on the OMM via the chaperone glucose-regulated protein 75 (GRP75)[45]. On the IMM, mitochondrial

calcium uniporter (MCU), a low affinity and high-capacity Ca^{2+} -selective ion channel, mediates Ca^{2+} entry into the mitochondrial matrix[46]. The low affinity of the MCU is compensated by the quasi-synaptic Ca^{2+} transfer of the IP3R-GRP75-VDAC1 complex[47].

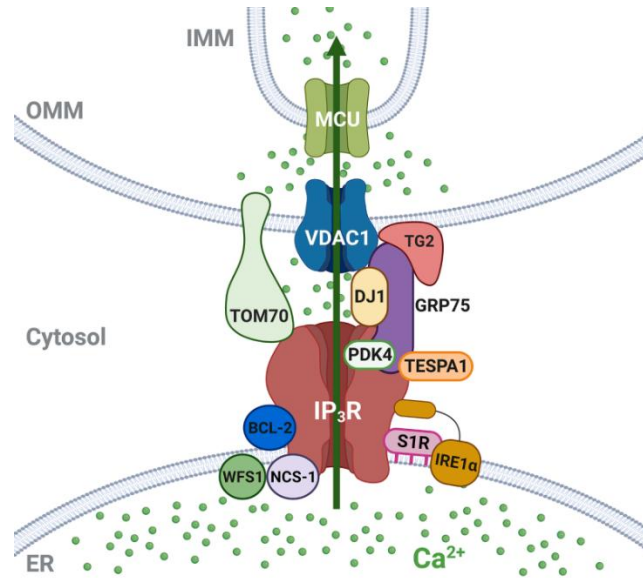


Figure 6. IP3R-GRP75-VDAC1 MAM complex and MCU surrounded by several regulators: WFS1; NCS1; BCL-2; S1R; IRE1α; TESPA1; PDK4; DJ1; TOM70; TG2. De Ridder et al., Cell Calcium, 2023.

Another important class of intracellular Ca^{2+} release channels are the ryanodine receptors (RyRs), which are abundantly expressed on the sarco/endoplasmic reticulum (S/ER) of excitable cells[48, 49]. Although the presence of RyRs at MAMs is less established than that of IP3Rs, they similarly contribute to the amplification of Ca^{2+} signals through Ca^{2+} -induced Ca^{2+} release, thereby generating robust intracellular Ca^{2+} responses[48, 49].

2.5 Ryanodine receptor

Ryanodine receptors (RyRs) are the largest known ion channels, and they owe their name to the ability to strongly bind the alkaloid plant ryanodine, which locks them into a subconductance state at micromolar concentrations and blocks them at higher concentrations[50].

They are expressed in excitable tissues and located on the sarco/endoplasmic reticulum where they are responsible for Ca^{2+} release[48]. Three human isoforms are described (RyR1, RyR2 and RyR3) by the literature, which share 65%

homology and show different levels of expressions depending on the cell type[50].

The RyR1 gene is located on the chromosome 19q13.2 and it is predominantly expressed in the skeletal muscle[51]. RyR2, also located on chromosome 19q13.2, was first described in the cardiac muscle[52]. Finally, RyR3, on chromosome 15q13.3-14, is mainly found in the brain and is also the least studied isoform[53].

X-ray crystallography and Cryo-EM studies have described the enormous structure of the RyR [Fig. 6][54, 48]. It is an homotetramer formed by four 565 kDa (~5000 amino acids) protomers which give to the channel a “mushroom” shape[55]. The N-terminal cytosolic part accounts for 90% of the receptor mass, while the small C-terminal transmembrane region accounts for the remaining 10%[55].

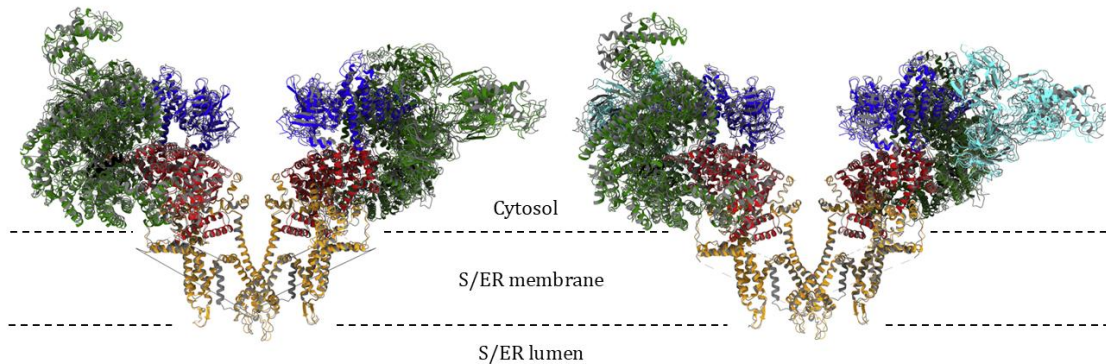


Figure 7. Side representation of RyR1 structure, with ribbon diagrams, in the primed (left) and open (right). Cytoplasmic shell and NTD are coloured in green. Cytoplasmic helical domains are coloured in blue. Regulatory regions are coloured in red. Pore domain and CTD are coloured in yellow. Figure adapted from Smith et al., *J. Structural Biology*, 2021.

Three principal domains of the RyRs have been reported. The pore domain is fundamental for ion conduction, it is formed by 6 transmembrane segments (6TM) and contains a pseudo-voltage sensor domain (pVSD) which can modulate the pore gating[48]. The activation domain contains a thumb and forefingers (TaF) domain, which clamps the zinc finger-containing C-terminal domain (CTD) providing the allosteric coupling between the pore and the shell of the receptor. Additionally, the activation domain presents a EF-hand motif that interacts with the pVSD and modulates the activation through Ca^{2+} levels[48]. The third and

largest domain is the cytoplasmic shell, which is mainly composed by α – solenoid repeats[48].

Different RyR modulation factors have been outlined. Firstly, as a Ca^{2+} induced Ca^{2+} release sensitive receptor, low quantities of Ca^{2+} activate the receptor while high quantities inhibit it by binding specific high - affinity and less selective low - affinity Ca^{2+} sites respectively[56, 57]. Furthermore, RyR1 in skeletal muscle cells and RyR2 in cardiac cells, because of their Ca^{2+} sensitivity, are regulated by voltage dependent Ca^{2+} channels (CaV)[49]. Additionally, Ca^{2+} can indirectly interfere with RyR activity through Ca^{2+} binding proteins calmodulin and calsequestrin, which bind directly to the cytosolic and intraluminal regions of the receptor, respectively[49].

Although Ca^{2+} has a bell shaped modulation on the RyRs, it appears to have a synergic effect when ATP is present[48]. ATP is indeed a well characterized physiological activating ligand for RyR. In this sense, the effect of caffeine is even stronger, as this exogenous ligand further increases the open probability[48].

Among negative modulators of RyR are Mg^{2+} ions and calstabin [49, 50]. The former inhibits the receptor by competing with Ca^{2+} for the specific high – affinity and less selective low affinity sites, while the latter ones bind directly to specific sites on the RyR, stabilizing its closed state[56]. Dantrolene, instead, is the only exogenous inhibitor clinically approved for malignant hypothermia, a condition in which a leaky mutated RyR1s cause uncontrolled muscle contraction[58].

RyRs are also involved in other diseases such as neurodegenerative disorders, cardiac diseases and metabolic disorders (diabetes mellitus) which are also clinical features to Wolfram syndrome[50]. Furthermore, considering that alterations in ROS levels can lead to leaky RyRs, and given the role of C1SD2 in maintaining calcium and ROS homeostasis, it is plausible to hypothesize a critical change in the interaction between these two factors in the manifestation of Wolfram syndrome type 2[21].

3. Aims of the internship activity

3.1 RyR3 and CISO2 interaction

The first goal of the internship activity was to assess a physical interaction between RyR3 and CISO2. This point was reached through a Co-Immunoprecipitation assay using anti RyR3 antibodies and detecting both RyR3 and CISO2 through immunoblotting.

3.2 Impact of CISO2 on RyR3 Ca²⁺ release

In a second moment a possible modulatory effect by CISO2 on the caffeine mediated activation of RyR3 was studied, the RyR3-evoked Ca²⁺ release from the ER, with FURA-2 fluorophore through fluorescence microscopy, in presence and absence of CISO2. The cells knocked out for CISO2 were created using CRISPR-Cas9 genome editing technology.

3.3 Impact of CISO2 mutants on RyR3 Ca²⁺ release

In addition, the impact of CISO2 mutations on RyR3-mediated Ca²⁺ flux was examined not only for the wild-type form of CISO2 but also for two variants: CISO2 FeS Cluster Disruption (CISO2^{FeSCD}) and CISO2^{N72S}. For these experiments, CISO2 knockout cells were deliberately transfected with mCherry vectors containing the mutant forms.

3.4 ER Ca²⁺ storage and mitochondrial Ca²⁺ studies

To further investigate Ca²⁺ dynamics, additional microscopy experiments were performed in the presence and absence of CISO2. In particular, ER Ca²⁺ storage was evaluated by Ca²⁺ leak in the cytosol after inhibition of SERCA2 with thapsigargin, while mitochondrial Ca²⁺ dynamics following RyR3 caffeine stimulation were visualized using the mtCEPIA fluorophore (mitochondrial Calcium-measuring Enhanced Protein Indicator with Autofluorescence).

4. Materials and methods

4.1 Antibodies

Co-Immunoprecipitation antibodies: monoclonal Anti-RyR3 C34 mouse antibody (Developmental Studies Hybridoma Bank, University of Iowa, Iowa, USA); nonspecific mouse IgGs (Santa Cruz Biotechnology, Cat# sc-2025, Heidelberg, Germany)

Primary immunoblot antibodies: monoclonal Anti-RyR3 C34 mouse antibody (1:1000 final conc.) (Developmental Studies Hybridoma Bank, University of Iowa, Iowa, USA); polyclonal Anti-CISD2 rabbit antibody (1:1000 final conc.) (ABclonal, Cat# A5231, Woburn, Massachusetts, USA); monoclonal Anti-Vinculin mouse antibody (1:1000 final conc.) (Sigma-Aldrich, Cat# V9131, Bornem, Belgium); monoclonal Anti-Flag M2 mouse antibody (1:1000 final conc.) (Sigma-Aldrich, Cat# F3165, Bornem, Belgium).

Secondary immunoblot antibodies: Pierce® Goat Anti-Mouse IgG peroxidase conjugated antibody (1:1000 final conc.) (Thermo Fischer, Cat# 31430, Merelbeke, Belgium); Anti-Rabbit HRP linked antibody (1:2000 final conc.) (Cell Signaling Technology, Cat# 7074, Danvers, Massachusetts, USA).

4.2 Plasmids

pSpCas9(BB)-2A-Puro (PX459) gRNA CISD2 plasmid, derived from *Streptococcus pyogenes*, was used for CRISPR-Cas9-mediated genome editing on Hek293 RyR3 cells. The plasmid contains a codon-optimized Cas9 nuclease, a 2A self-cleaving peptide for bicistronic expression, a puromycin resistance gene for antibiotic selection, and a single guide RNA (sgRNA) targeting the CISD2 gene. Guide RNA sequence (GGAGCTGCACCTTCACGATA) was obtained from Synthego (Redwood City, California, USA). The plasmid was created using the PX459 vector backbone, which is available from Addgene (Feng Zhang Addgene plasmid #48139; <http://n2t.net/addgene:48139>; RRID: Addgene_48139).

pCMV26 and pCMV24-p2A-mCherry empty vectors; pCMV26-CISD2 and pCMV24-CISD2-p2A-mCherry CISD2 wild type vectors; pCMV26-CISD2-

CISD2^{FeSCD} and pCMV24-CISD2^{FeSCD}-p2A-mCherry CISD2 FeS Cluster Disruption (FeSCD) form vectors; pCMV26-CISD2^{N72S} and pCMV24-CISD2^{N72S}-mCherry CISD2^{N72S} form vectors.

4.3 Cell culture

All the mentioned media and supplements added to the medium were purchased from Life Technologies (Ghent, Belgium) unless stated otherwise.

HEK293 RyR3 cells were cultured in 500 mL of minimum essential medium (MEM) Alpha Medium 1X; 1:10 of fetal bovine serum (FBS) Standard; 1:100 of GlutaMAX® (100X); 1:50 of Penicillin Streptomycin (Pen Strep) antibiotic mixture; 1:125 of selective reagent G418 (800 µg/µL) (InvivoGen, San Diego, California, United States).

HEK RyR3 cells were preserved in liquid nitrogen with serum and DMSO, then suspended with 4 mL of medium and centrifuged at 300 G at room temperature for 5 minutes. The obtained cell pellet was resuspended in 10 mL of medium and added in a Falcon tube in a total volume of 30 mL of medium. The entire volume containing HEK RyR3 cells was seeded in a 75 cm² flask (SARSTEDT AG & Co. KG, Nümbrecht, Germany) and grown at 37°C and 10% CO₂.

Cells were regularly passaged upon reaching near-confluency. HEK RyR3 cells in the flask were detached using 5 mL of Versene 1:5000 1X (0.2 g EDTA(Na₄) per Liter of Phosphate Buffered Saline (PBS)), 3 mL of 0,05% Trypsin EDTA 1X (0.5 g/L of trypsin (1:250) and 0.2 g/L of EDTA4Na) at 37°C. Subsequently, 12 mL of MEM Alpha Medium 1X were added to stop trypsinization. Cell suspension was counted using a Countess cell counter (Life Technologies, Carlsbad, California, United States). 10 µL of the cell sample was added to 10 µL Trypan Blue (Thermo Fisher Scientific, Waltham, Massachusetts, United States) to discriminate between live and dead cells during the count.

4.4 Cell transfection and selection

All the mentioned media and supplements added to the medium were purchased from Life Technologies (Ghent, Belgium) unless stated otherwise.

HEK293 RyR3 cells were seeded in CELLSTAR® 6-well cell culture plates (Greiner Bio-One, Kremsmünster, Austria) at a density of 7×10^5 cells per well in 200 μ L minimum essential medium (MEM) Alpha Medium already containing FBS, GlutaMAX®, Pen Strep and G418. Cells were transfected with the plasmid pS-Cas9-2A-puro-PX459 (Addgene #48139), either containing a guide RNA (gRNA) targeting the C1SD2 gene to generate HEK293 RyR3 C1SD2 knockout (KO) cells, or without the gRNA to generate wild-type (WT) control cells.

Transfection was performed using TransIT®-X2 transfection reagent (Mirus Bio, Madison, WI, USA) at a 2:1 reagent-to-DNA ratio (μ L/ μ g). The plasmids and the transfection reagent were incubated together for 20 min at room temperature in 200 μ L of Dulbecco's Modified Eagle's Medium (DMEM) without supplements to allow complex formation. The medium was then added to each well, and cells were incubated for 24 hours, after which medium was refreshed.

In the case of selection, cells were selected with 3 μ g/mL puromycin (InvivoGen, San Diego, CA, USA) for 48 hours. Successful C1SD2 knockout was subsequently confirmed by immunoblotting.

4.5 Bradford protein assay

Unless specifically stated, all chemicals and consumables were obtained from Thermo Fischer (Merelbeke, Belgium).

HEK RyR3 cells were lysed with 400 μ L 3-((3-cholamidopropyl) dimethylammonio)-1-propanesulfonate (CHAPS)-based lysis buffer (pH 7.5, 50 mM Tris-HCl, 100 mM NaCl, 2 mM EDTA, 50 mM NaF, 1 mM Na₃VO₄, 1% CHAPS and protease inhibitor tablets (Roche, Basel, Switzerland)) for 30 minutes in ice and rapidly vortexed every 10 minutes. The lysates were centrifuged at 5000 G, 4 °C for 5 minutes and the supernatant was collected. The Bradford assay's samples derived from these lysates.

6 Albumin Standards were assembled with 6 different concentrations of bovine serum albumin (BSA) (0 μ g/mL; 100 μ g/mL; 200 μ g/mL; 400 μ g/mL; 800 μ g/mL; 1600 μ g/mL), a fixed concentration of 10% azide solution and Milli-Q water in a total volume of 1500 μ L.

Sample and BSA standards were pipetted in a CELLSTAR® 96 Well Cell Culture Plate (Greiner Bio-One, Kremsmünster, Austria) with a volume of 200 µL of Bradford Dye Reagent Ready-to-Use solution. Protein concentrations and standard curve were subsequently measured by FlexStation® 3 Multi-Mode Microplate Reader (Molecular Devices, San Jose, California, United States). The back calculation from optical densities to protein standard curve was performed by SoftMax Pro® software 4 Parameter equation ($y = (A - D)/(1 + (x/C)^B) + D$).

4.6 Co-Immunoprecipitation (Co-IP)

Unless specifically stated, all chemicals and consumables were obtained from Thermo Fischer (Merelbeke, Belgium).

For the Co-IP assay, HEK RyR3 cells were cultured in medium size flask in MEM Alpha (minimum essential medium) for 2 days and then passaged. After the cell count, an average concentration of 5.2×10^5 cells/mL was estimated. HEK RyR3 cells were lysed with 400 µL CHAPS-based lysis buffer (pH 7.5, 50 mM Tris-HCl, 100 mM NaCl, 2 mM EDTA, 50 mM NaF, 1 mM Na₃VO₄, 1% CHAPS and protease inhibitor tablets (Roche, Basel, Switzerland)) and vortexing.

First, 20 µL of Dynabeads® Protein G was added and beads were washed with PBS (Phosphate Buffered Saline) + 0.02% Tween. The solution was then removed after separating Dynabeads® Protein G from the solution on a magnet. 5 µg of RyR3 C34 mouse antibody (Developmental Studies Hybridoma Bank, University of Iowa, Iowa, USA) and 5 µg of aspecific mouse IgGs (Santa Cruz Biotechnology, Heidelberg, Germany) was added to the Dynabeads® Protein G in two separated Eppendorf tubes and incubated for 20 minutes at room temperature on a head-over-head rotator.

The supernatant was removed on magnet while the remaining pellet was washed with 200 µl PBS + 0.02% Tween, then 400 µg HEK RyR3 lysate, in a total volume of 500 µl CHAPS buffer, were added. HEK RyR3 lysate, RyR3 C34 mouse antibodies, Dynabeads® Protein G and HEK RyR3 lysate, nonspecific mouse IgGs, Dynabeads® Protein G incubated overnight at 4°C on a head-over-head rotator.

The beads were washed five times with 200 μ L CHAPS buffer to reduce nonspecific bindings and protein-protein interactions. After the fifth wash, samples were eluted in 50 μ L of elution solution (SDS 0.2% + tween 0.1% + tris 50 mM pH 8) for 30 minutes in a shaker at 1000 rpm, room temperature. Subsequently, eluted samples were collected on magnet and separately combined with 16 μ L of stabilization buffer NuPAGE® LDS Sample Buffer 4X (988 mM Tris, 2.04 mM EDTA, 8 % LDS (Lithium Dodecyl Sulfate), 40 % Glycerol, 0.88 % Commassie Brilliant Blue G250, 0.7 mM Phenol red).

4.7 SDS-PAGE electrophoresis and immunoblotting

Unless specifically stated, all chemicals and consumables were obtained from Thermo Fischer (Merelbeke, Belgium).

NuPAGE® LDS Sample Buffer containing 1:200 99% extra pure 2-mercaptoethanol (Acros Organics, Geel, Belgium) was added to all immunoblot samples which, following heating for 5 minutes at 90°C, were run on NuPAGE® 4–12 % Bis-Tris gels in MES running buffer (MES (50 mM), Tris base (50 mM), SDS (0,1%), and EDTA (1 mM), pH of 7.3) at 180V for 45 minutes.

Subsequently, the samples were transferred on a Immobilon®-FL Transfer Membrane (Merck Millipore, Hoeilaart, Belgium) pre-activated in methanol in 10x Tris/Glycine Buffer (25 mM Tris, 192 mM glycine, pH 8.3) (Bio-Rad®, Lokeren, Belgium) at 100V for 60 minutes.

The membrane was blocked with a solution of 5% milk in Tris-buffered saline (TBS) with 0.1% Tween® for 60 minutes followed by overnight incubation in TBS – 0.1% Tween solution with 1% milk with the respective primary antibodies on a rotor at 4°C.

After the incubation, the membrane was washed three times for 5 minutes in TBS – 0.1% Tween and then incubated in TBS – 0.1% Tween solution with 1% milk with the respective secondary antibodies on a rotor for 45 minutes at room temperature.

Afterward, the membrane was washed three times for 5 minutes each with TBS containing 0.1% Tween-20. The Pierce® ECL Western Blotting Substrate was then applied to the membrane prior to signal detection. Band visualization was subsequently performed using the Bio-Rad ChemiDoc™ MP Imaging System, which enabled both colorimetric detection and high-resolution, high-sensitivity chemiluminescent imaging at various exposure times.

4.8 Single cell ratiometric Ca²⁺ imaging

For FURA-2 measurements, a Nikon TI2-E inverted microscope equipped with a 20 × 0.5 NA Plan Fluor DIC N2 air objective and a pco. edge 4.2bi sCMOS camera was used. FURA-2 was alternately excited at a 2 s interval for caffeine response and at a 10 s interval during thapsigargin response using a CoolLED pE-300 ultra/pE-340 lamp set at 340 nm and 380 nm (CoolLED, Andover, UK), and using a dichroic mirror FF02-409/LP-25 (Semrock, New York, USA) and a bandpass emission filter 515/30 (Semrock, Rochester, USA). mCherry and Green mtCEPIA were visualized using a CoolLED pR-4000 lamp (CoolLED, Andover, UK) at 550 nm and 470 nm respectively, using a CoolLED pR-4000 lamp, and using the cubical excitation filterset FF01-378/474/554/635 and bandpass emission filters of 595/31 and 515/30, respectively. (Semrock, Rochester, USA).

4.9 Single cell caffeine response

Both Hek RyR3 wild type and C1SD2 KO line cells were seeded in 4 chamber glass bottom dishes with coverslips (IBL, Gerasdorf bei Wien, Austria). For mCherry microscopy assay, C1SD2 KO cells were transfected with 75 ng of empty vector DNA and 150 ng of C1SD2 wild type and mutant vectors. For mitochondrial calcium measurements, both HEK RyR3 wild-type and C1SD2 KO cells were transfected with 100 ng of Green mtCEPIA. In both transfection conditions, TransIT®-X2 transfection reagent was used, followed by a 2-day incubation period. On the day of the microscopy analysis the cells were washed once with 1.5 mM Ca²⁺ Krebs buffer (135 mM NaCl, 6.2 mM KCl, 1.2 mM MgCl₂, 12 mM HEPES, pH 7.3, 12 mM glucose and 1.5 mM CaCl₂) and incubated with 1:1000 FURA-2-AM (AnaSpec, Fremont, CA, USA) in 1.5 mM Ca²⁺ Krebs for 30 minutes, in the dark at room temperature. After the incubation the cells were washed twice again with

1.5 mM Ca^{2+} Krebs buffer and allowed to rest for 30 minutes, in the dark at room temperature. In the meantime, 3 mM 3 mM ethylene glycol-bis(β -aminoethylether)-N,N,N',N'-tetraacetic acid (EGTA) Ca^{2+} chelation solution, 1.5 mM caffeine and 3 mM caffeine were prepared in 0 mM Ca^{2+} Krebs buffer (135 mM NaCl, 6.2 mM KCl, 1.2 mM MgCl_2 , 12 mM HEPES, pH 7.3 and 12 mM glucose) already containing a concentration of 3 mM EGTA. In both the presence and absence of the mCherry condition, measurements were performed by adding 1 mL of 3 mM EGTA after 30 seconds, followed by 1 mL of 1.5 mM caffeine after 1 minute, and 1 mL of 3 mM caffeine after an additional 2 minutes. Conversely, in the mtCEPIA condition, 1 mL of 2 mM ionomycin, prepared in 0 mM Ca^{2+} Krebs buffer containing 3 mM EGTA, was added after 3 minutes instead of 3 mM caffeine as a positive control.

4.10 Single cell thapsigargin response

Both Hek RyR3 wild type and C1SD2 KO line cells were seeded in 4 chamber glass bottom dishes with coverslips. On the day of fluorescence measurement, the cells were treated as described in “Single cell caffeine response” section. The measurements were performed by adding 1 mL of 3 mM EGTA after 30 seconds, followed by 1 mL of 2 μM thapsigargin solution, previously prepared in 0 mM Ca^{2+} Krebs buffer containing 3 mM EGTA, after 1 minute. Upon thapsigargin addition, the imaging interval was changed from 2 seconds to 10 seconds to prevent cell damage and was maintained for 8 minutes.

In both the caffeine and thapsigargin responses, all solutions were added in excess. Furthermore, during the measurements, continuous suction through a needle was applied in order to remove the excess solution.

4.11 Single cell ratiometric Ca^{2+} imaging data analysis

Data was quantified using a custom ImageJ script, which can be found at https://github.com/jensloncke/ImageJ_macros/blob/master/FURA/FURA.ijm (for not mCherry transfected cell measurements) and https://github.com/jensloncke/ImageJ_macros/blob/master/FURA%20transfected%20cells/FURA_transfection_HEK.ijm (for mCherry transfected cell

measurements). Traces of the ER Ca²⁺ flux and relevant response parameters, such as area under curve or maximum amplitude, were extracted using custom Python scripts with numpy[59], pandas[60], and plotly packages: https://github.com/jensloncke/LMCS-python-scripts/blob/main/Extract%20multi%20measure%20data/data_extraction_multi_measure_image.py (traces); https://github.com/jensloncke/LMCS-python-scripts/blob/main/Quantify%20agonist%20response/quantify_response.py (relevant response parameters). All the represented plots were made using ggplot2[61] package in the R programming language[62]. In order to perform a statistical analysis of the differences in the thapsigargin and caffeine response groups assumptions of normal distribution of residuals and equality of variances was checked via Shapiro-Wilk testing and Levene testing, respectively. In the event that assumptions were met, either a two-sample independent *t*-test or a two-way ANOVA followed by Tukey's HSD post hoc test was applied. Conversely, when either the assumption of normality or homogeneity of variances was violated, nonparametric tests such as the Wilcoxon rank-sum test or the Kruskal-Wallis's test followed by Dunn's post hoc test with Holm-Bonferroni correction were performed. Two-way ANOVA was applied using the car package[63] while two samples *t*-test, Tukey's HSD post-hoc tests, Wilcoxon rank-sum test, Kruskal-Wallis tests and Dunn's post hoc tests were applied using R stats package.

5. Results and conclusions

5.1 CISO2 physically interacts with RyR3

The interaction between CISO2 and RyR3 was investigated using Co-Immunoprecipitation (Co-IP) assays. As described in the “Materials and methods” section, RyR3 was precipitated using anti-RyR3 C34 antibodies coupled to magnetic beads. The Co-IP results were analysed by SDS-PAGE electrophoresis followed by immunoblotting.

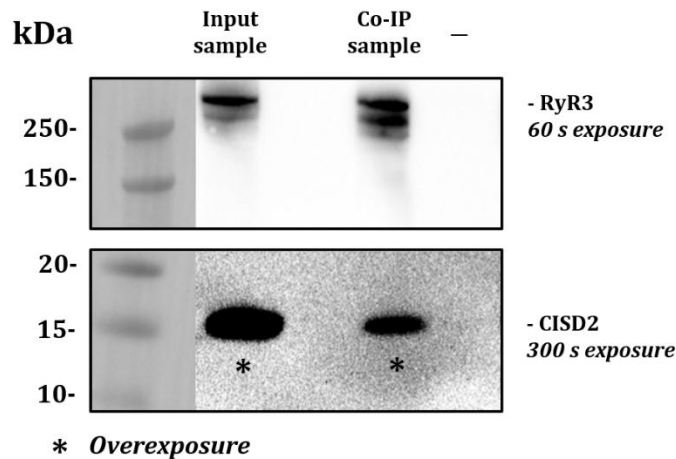


Figure 8. Immunoblot of co-immunoprecipitation experiment using anti-RyR3 C34 and anti-CISO2 antibodies. RyR3 bands detected with a 60 second exposure and CISO2 bands detected with a 300 second exposure in both input and Co-IP lanes. CISO2 bands appear overexposed. The negative control lane (-) does not present any band. The immunoblot shown is a representative blot out of three biological replicates ($N = 3$).

Immunoblotting, performed using the Bio-Rad ChemiDoc™ MP Imaging System, revealed a band above 250 kDa, consistent with the expected molecular weight of RyR3 (565 kDa), in both the input sample (used as a positive control) and the Co-IP sample. Additionally, a band around 15 kDa, corresponding to CISO2 (15.3 kDa), was observed in these two lanes [Fig. 8].

Notably, both the Co-IP and input lanes showed the presence of RyR3 and CISO2. This indicates that CISO2 co-precipitated with RyR3 during the Co-IP assay, and the detected band likely represents the fraction of CISO2 physically associated with RyR3. These results support a direct physical interaction between CISO2 and RyR3 in cells expressing RyR3, such as the HEK293 RyR3 cells used in this study.

5.2 Generation of HEK293 RyR3 C1SD2 knockout cells

In order to examine the impact of C1SD2 in RyR3-mediated Ca^{2+} release, HEK293 RyR3 C1SD2 knockout (KO) cells were generated using CRISPR-Cas9 technology.

The efficiency of the knockout was evaluated by immunoblotting after SDS-PAGE, with vinculin detected by specific antibodies and used as a loading control in all samples.

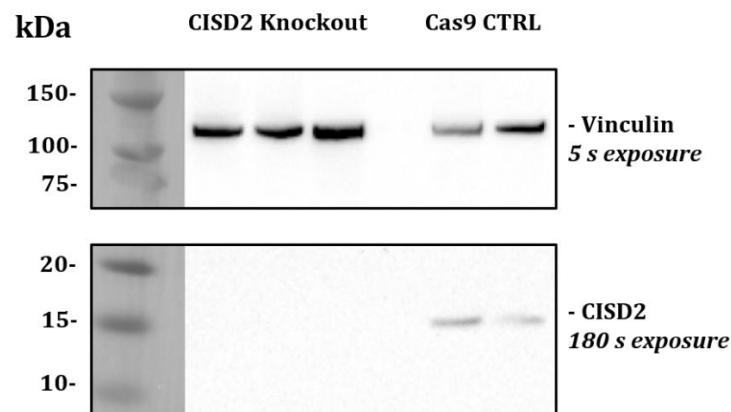


Figure 9. Immunoblot of HEK293 cells transfected with pSpCas9(BB)-2A-Puro (PX459) plasmids containing either C1SD2-targeting gRNA or an empty vector (control). Vinculin bands were detected after a 5 second exposure in all samples, whereas C1SD2 bands were detected after a 180 second exposure only in the control (Cas9 CTRL) samples. To increase the likelihood of successful knockout, the experiment was performed in triplicate, and all replicates were successful.

As shown in the previous blot [Fig. 8], two distinct C1SD2 bands were detected in the Cas9 control lanes (Cas9 CTRL), which correspond to lysates from cells transfected with the Cas9 plasmid without a C1SD2-specific guide RNA. In contrast, no C1SD2 signal was observed in the lanes corresponding to the C1SD2 knockout cells [Fig. 9]. In conclusion, all three knockout samples shown on the immunoblot were successfully knocked out for the C1SD2 gene.

5.3 C1SD2 modulates RyR3 mediated Ca^{2+} release

The ER Ca^{2+} flux mediated by RyR3 was evaluated using fluorescence microscopy in the presence or absence of C1SD2. HEK293 RyR3 (Cas9 CTRL) and HEK293 RyR3 C1SD2 KO (C1SD2 KO) cells were loaded with FURA-2-AM, followed by the addition of 3 mM EGTA to chelate extracellular Ca^{2+} . Subsequently, 1.5 mM caffeine was applied to stimulate RyR3-mediated Ca^{2+} release from the ER [Fig. 10].

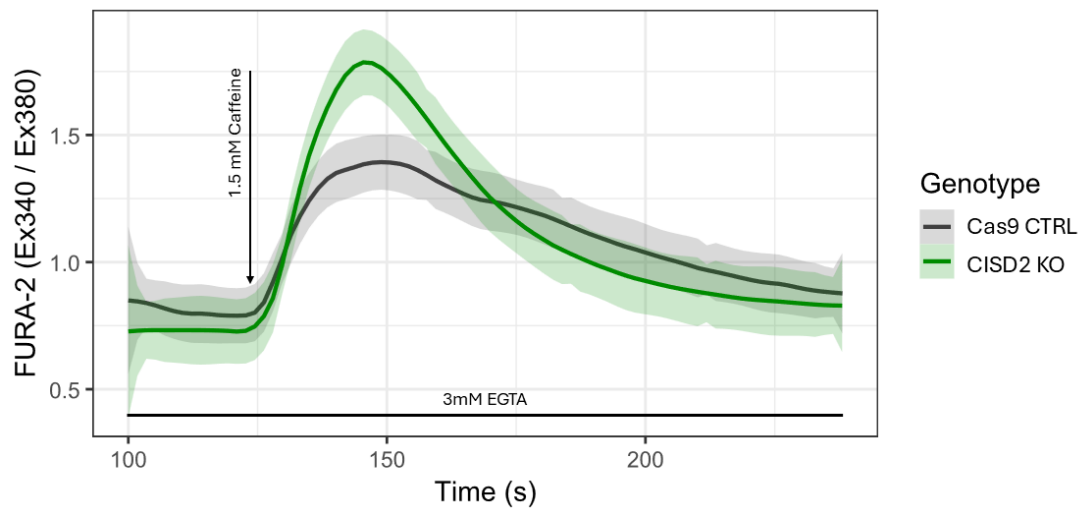
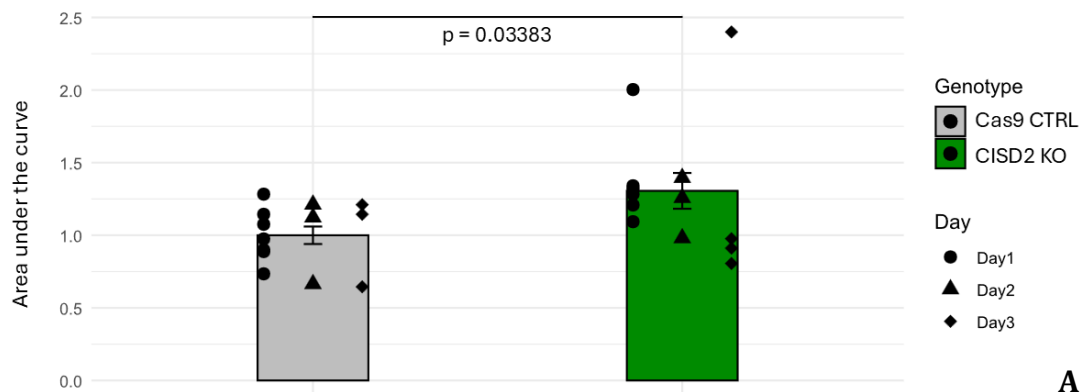


Figure 10. Traces of ER Ca^{2+} flux in HEK293 RyR3 cells (grey; Cas9 CTRL) and in HEK293 RyR3 C1SD2 KO cells (green, C1SD2 KO) after 1.5 caffeine mediated stimulation of RyR3. Values represent the ratio of Ca^{2+} -bound Fura-2 (excitation at 340 nm) to Ca^{2+} -free Fura-2 (excitation at 380 nm), measured in the presence of EGTA to chelate extracellular calcium.

The traces shown in Fig. 10 display a rapid increase in cytoplasmic Ca^{2+} following caffeine addition, followed by a gradual decline over time. This rise is clearly due to RyR3 activation on the ER membrane in response to caffeine. Furthermore, the prior addition of EGTA ensures that the Ca^{2+} reaching the cytosol originates from intracellular sources, and not from the extracellular environment.

Particularly, it is to be noted that there is a visible difference in the Ca^{2+} traces of cells expressing C1SD2 endogenously (Cas9 CTRL) and the cells without C1SD2 (C1SD2 KO). This difference will be exhaustively described in the following figures.



A

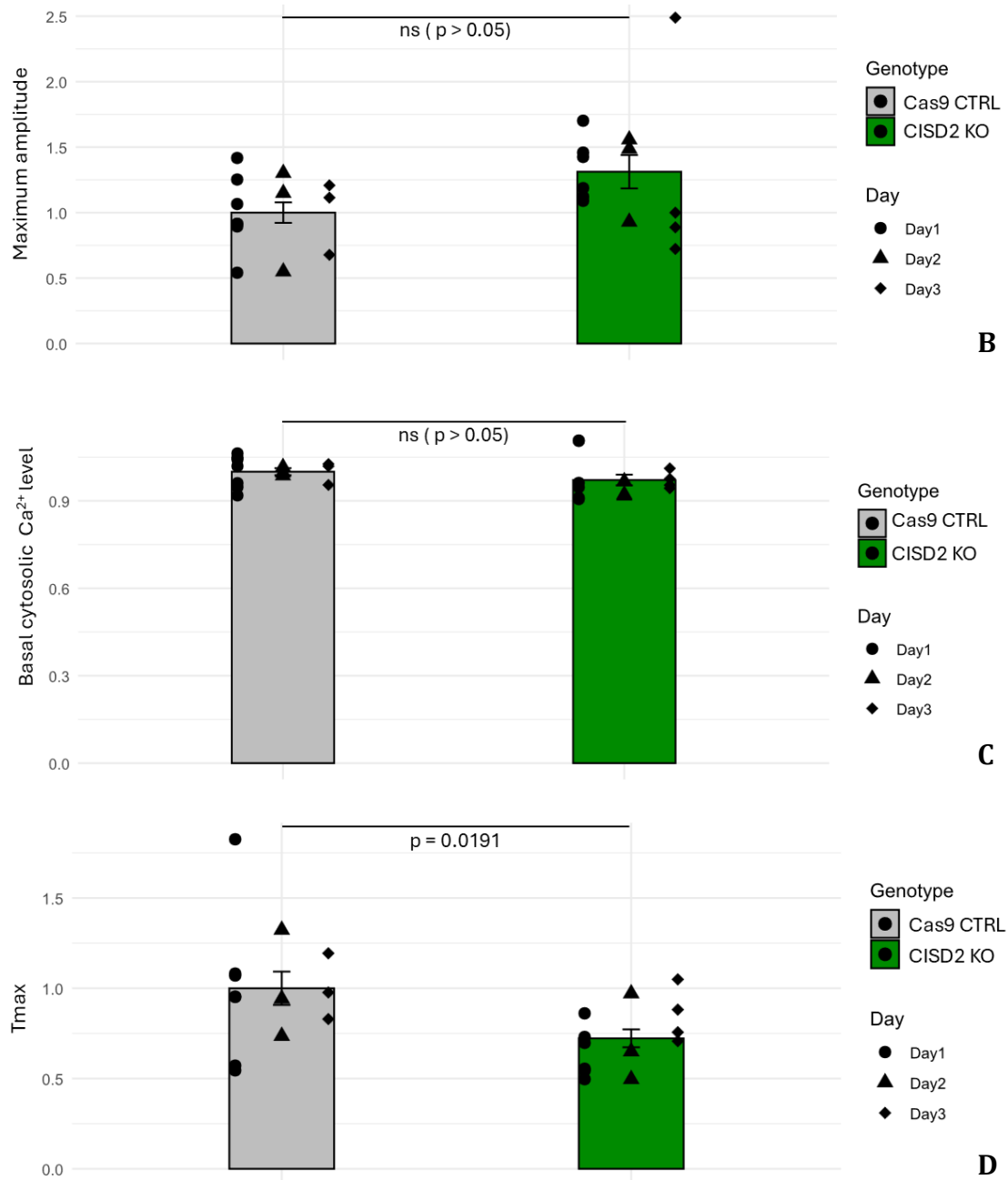


Figure 11 (A-D). Plots showing area under the curve (A), maximum amplitude (B), basal cytosolic Ca^{2+} level (C), and time to maximum (D). Data were obtained from ratiometric FURA-2 measurements of ER Ca^{2+} flux following caffeine stimulation of RyR3 in HEK293 RyR3 cells. Results are shown for C1SD2 KO (green) and C1SD2 WT (grey) conditions. Measurements were performed on three independent experimental days, indicated as Day 1 (circle), Day 2 (triangle), and Day 3 (square), which can be considered biological triplicates of the same experiment.

To better explain the differences between the two Ca^{2+} fluxes, several parameters were extracted from the measurements, including the area under the curve (AUC), the maximum amplitude, the basal cytosolic Ca^{2+} level, and the time to reach the maximum (Tmax).

The AUC is clearly greater in the C1SD2 KO condition [Fig. 11, A]. Although the maximum amplitude is also higher in the C1SD2 KO, this difference is not statistically significant [Fig. 11, B]. Similarly, the difference in basal cytosolic Ca^{2+} levels, measured before the addition of EGTA and caffeine, is not significant [Fig. 11, C]. Finally, a notable difference in T_{max} is observed [Fig. 11, D], indicating that Ca^{2+} reaches its maximum cytosolic concentration more slowly in the presence of C1SD2 compared to its absence.

Consequently, these results suggest that C1SD2 can directly modulate RyR3 by reducing the amount of Ca^{2+} released into the cytosol and slowing the activation of the channel. Importantly, this effect is independent of basal cytosolic Ca^{2+} levels, as no significant differences were observed. Therefore, C1SD2 acts as a negative regulator of RyR3.

5.4 Ca^{2+} flux differences are independent of ER Ca^{2+} storage

To further investigate the cause of the ER Ca^{2+} release difference between the two conditions, ER Ca^{2+} storage was also evaluated. Thapsigargin, a potent SERCA inhibitor, was applied to induce uncontrolled Ca^{2+} leakage from the ER into the cytosol, which was continuously monitored by fluorescence microscopy.

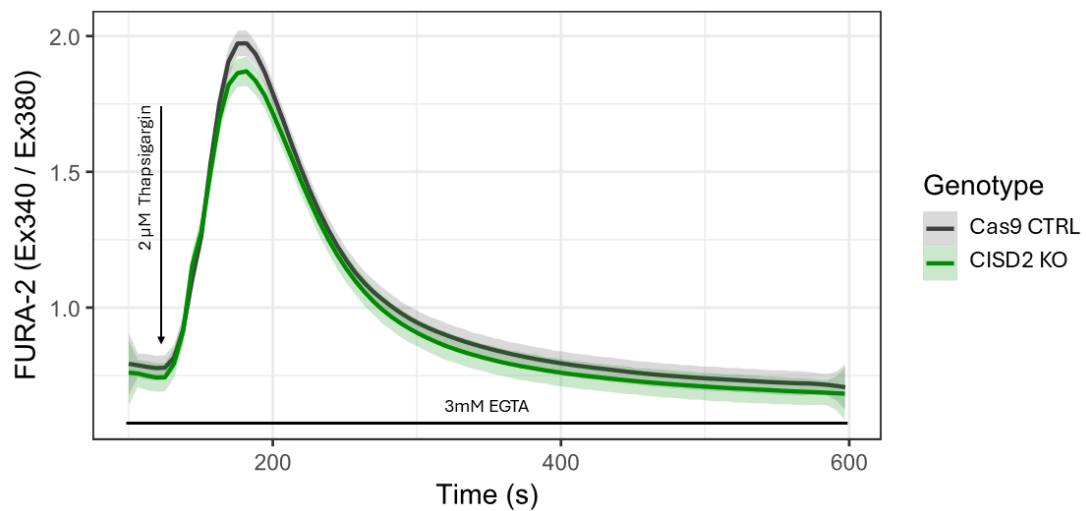


Figure 12. Traces of ER Ca^{2+} flux, representative of ER Ca^{2+} storage, in HEK293 RyR3 cells (grey; Cas9 CTRL) and in HEK293 RyR3 C1SD2 KO cells (green, C1SD2 KO) after 2 μM thapsigargin inhibition of SERCAs. Additional details are provided in Figure 9.

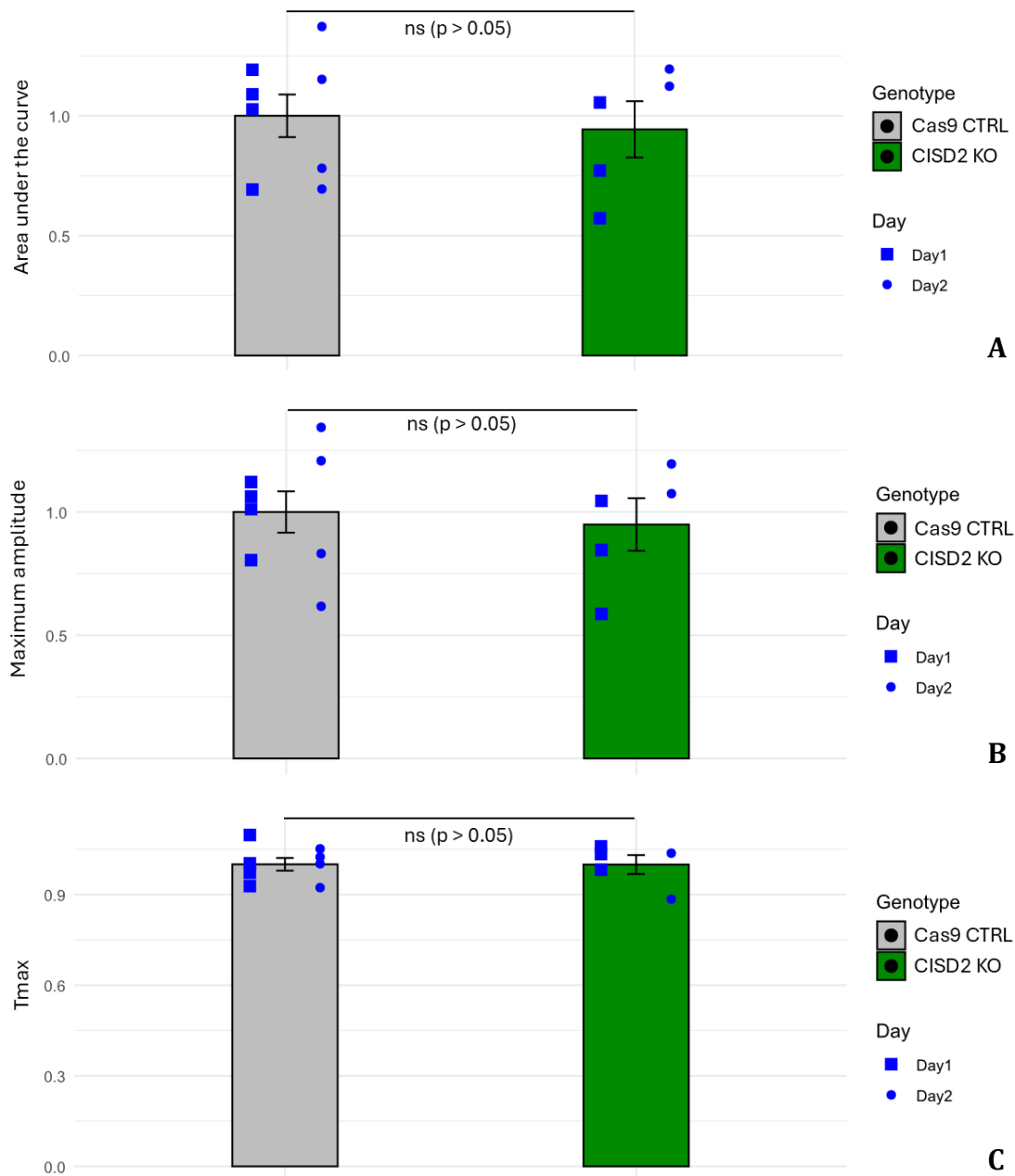


Figure 13 (A-C). Plots showing area under the curve (A), maximum amplitude (B), and time to maximum (C). Data were obtained from ratiometric FURA-2 measurements of ER Ca^{2+} flux following thapsigargin inhibition of SERCAs. Results are shown for Cisd2 KO (green) and Cisd2 WT (grey) conditions. Measurements were performed on two independent experimental days, indicated as Day 1 (square) and Day 2 (circle) which can be considered biological replicates.

ER Ca^{2+} storage traces [Fig. 12] indicate a very similar Ca^{2+} content in the endoplasmic reticulum under both Cisd2 present and absent conditions. This similarity is further confirmed by the quantitative analyses of AUC, maximum amplitude, and time to peak [Fig. 13, A-C], which reveal no significant differences

either in ER Ca^{2+} content [Fig. 13, A; B] or in the speed of Ca^{2+} release following thapsigargin inhibition [Fig. 13, C].

5.5 Impact of $\text{CISD2}^{\text{FeSCD}}$ and $\text{CISD2}^{\text{N72S}}$ mutants on RyR3 Ca^{2+} release

Subsequently, HEK293 RyR3 CISD2 knockout (KO) cells were transfected with mCherry vectors containing two different CISD2 mutant constructs in order to investigate their effects on RyR3. The mCherry tag allowed the identification and selection of correctly transfected cells prior to recording measurements. The first mutant, $\text{CISD2}^{\text{FeSCD}}$ (Fe–S cluster disruption), is an artificially generated variant of CISD2 carrying mutations in the four amino acids (Cys99, Cys101, Cys110, and His114) that coordinate the 2Fe–2S cluster [Fig. 3]. The second mutant, $\text{CISD2}^{\text{N72S}}$, results from a point mutation previously described as one of the five CISD2 variants associated with WS2, as discussed in section 2.3. In addition, a third mCherry vector containing CISD2 WT was used to assess rescue of the WT condition, while a fourth, empty mCherry vector served as a CISD2 KO control, whose behaviour had already been characterized in the previous experiment.

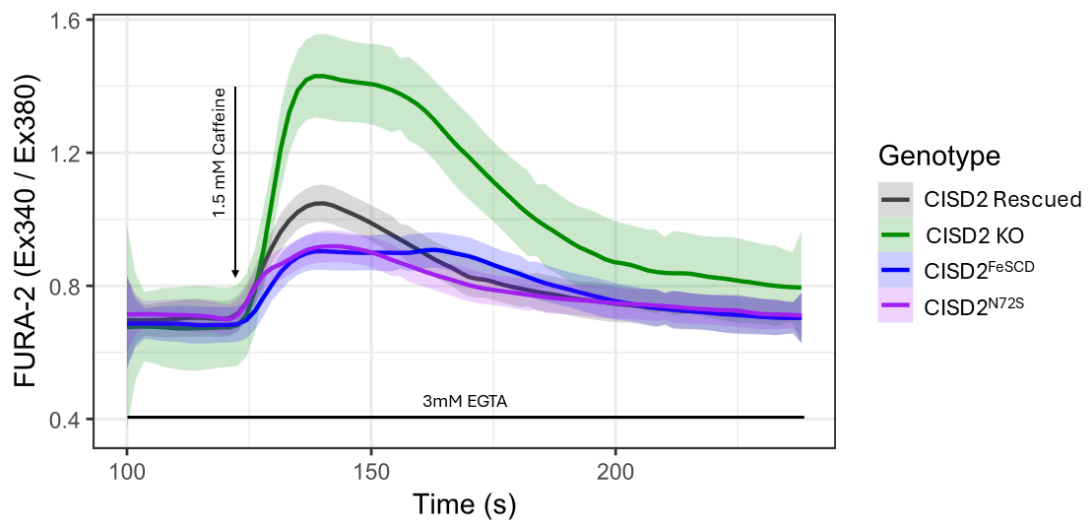
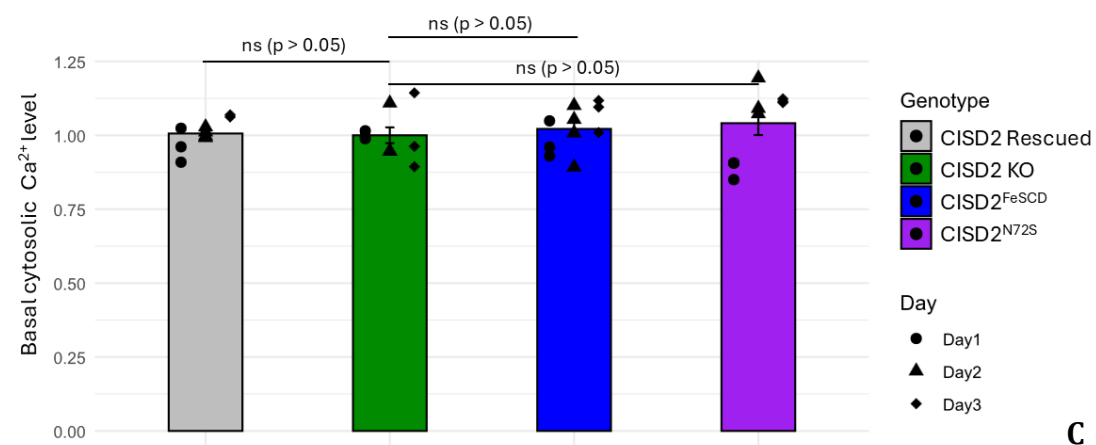
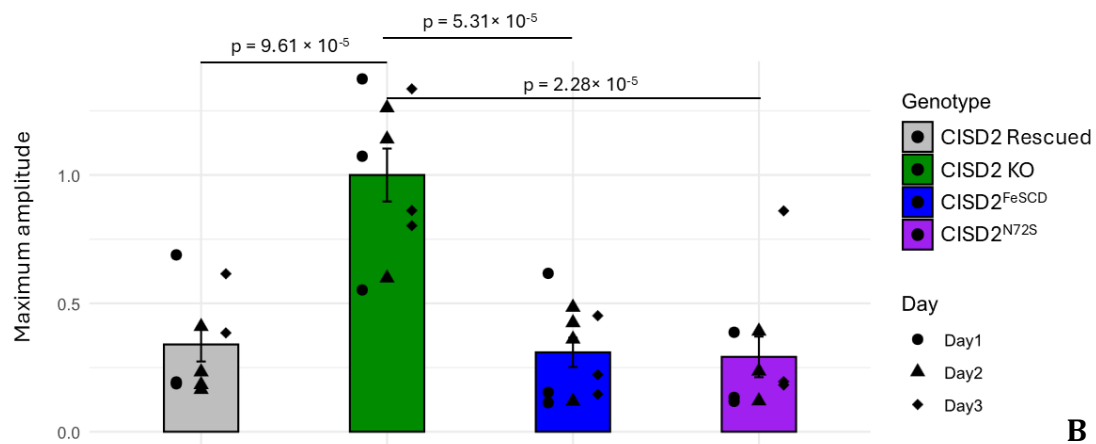
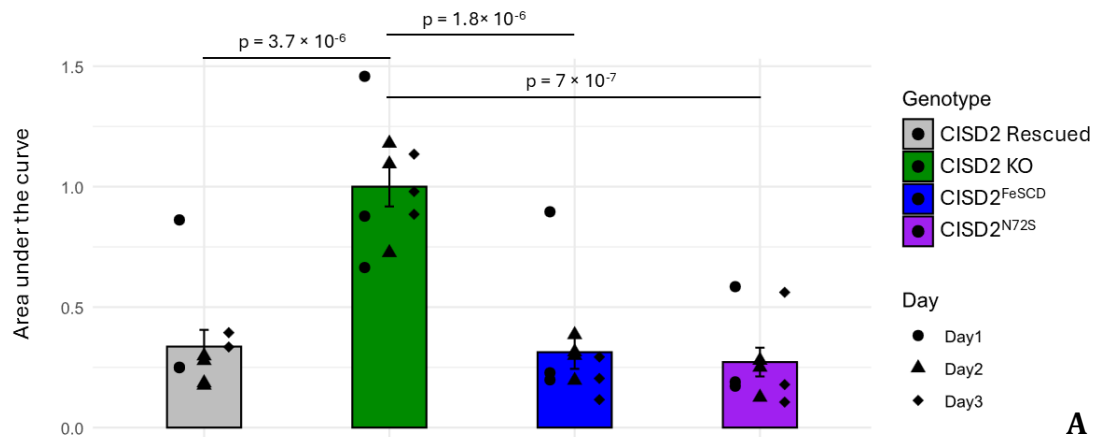


Figure 14. Traces of caffeine stimulated ER Ca^{2+} flux in HEK293 RyR3 CISD2 KO cells transfected with mCherry vectors containing CISD2 WT (grey; CISD2 Rescued), $\text{CISD2}^{\text{FeSCD}}$ (blue; $\text{CISD2}^{\text{FeSCD}}$), $\text{CISD2}^{\text{N72S}}$ (purple; $\text{CISD2}^{\text{N72S}}$) or an empty mCherry vector representing the CISD2 KO condition (green; CISD2 KO). Values were measured in presence of 3mM EGTA. Additional details are provided in Figure 9.

RyR3 Ca^{2+} release traces shown in Figure 14 exhibit the largest response in the **CISD2 KO** condition (green). In contrast, cells transfected with the wild-type form of CISD2 (grey) display reduced Ca^{2+} release, comparable to that observed in the

two mutant conditions (CISD2^{FeSCD}, blue; CISD2^{N72S}, purple). For a more precise evaluation of these results, the corresponding quantification plots should be considered [Fig. 15].



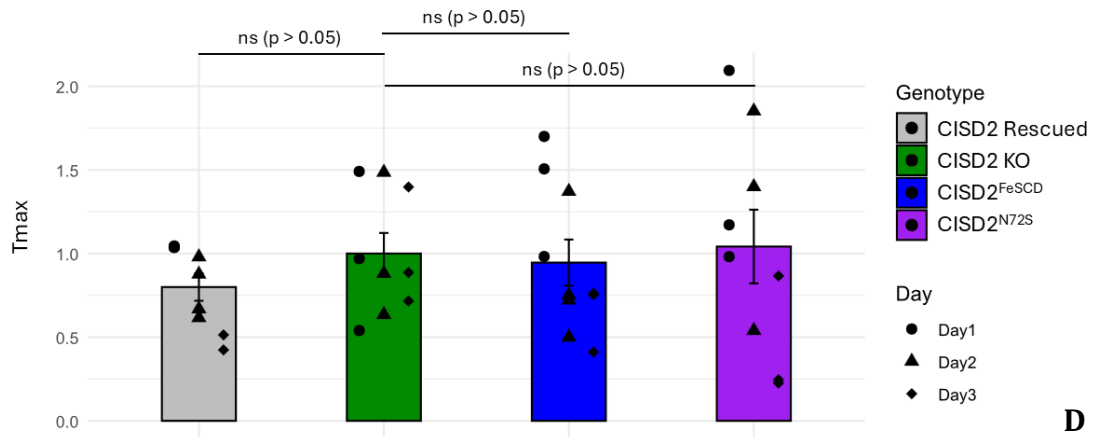


Figure 15 (A-D). Plots showing area under the curve (A), maximum amplitude (B), basal cytosolic Ca^{2+} level (C), and time to maximum (D). Data were obtained from ratiometric FURA-2 measurements of ER Ca^{2+} flux following caffeine stimulation of RyR3 in HEK293 RyR3 C1SD2 KO cells transfected with mCherry vectors. Results are presented for C1SD2 KO (green; empty mCherry), C1SD2 Rescued (grey; C1SD2 WT mCherry), C1SD2^{FeSCD} (blue; C1SD2^{FeSCD} mCherry), and C1SD2^{N72S} (purple; C1SD2^{N72S} mCherry) conditions. Measurements were performed on three independent experimental days, indicated as Day 1 (circle), Day 2 (triangle), and Day 3 (square), which can be considered biological triplicates of the same experiment.

AUC and maximum amplitude plots [Fig. 15, A; B] clearly reflect the enhanced Ca^{2+} release for the C1SD2 KO condition, already observed in the traces [Fig. 14]. This result reinforces the observation mentioned in the previous section regarding RyR3-mediated release in the presence and absence of C1SD2 [Fig. 10-11]. Moreover, cells transfected with the mCherry vector carrying the physiological form of C1SD2 exhibit a smaller response compared to the KO, suggesting a rescue of the WT phenotype through re-expression of C1SD2.

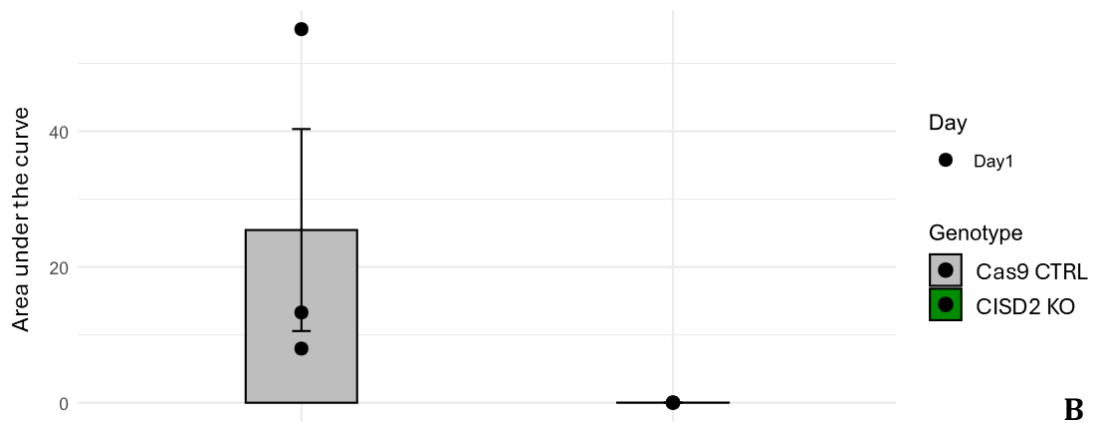
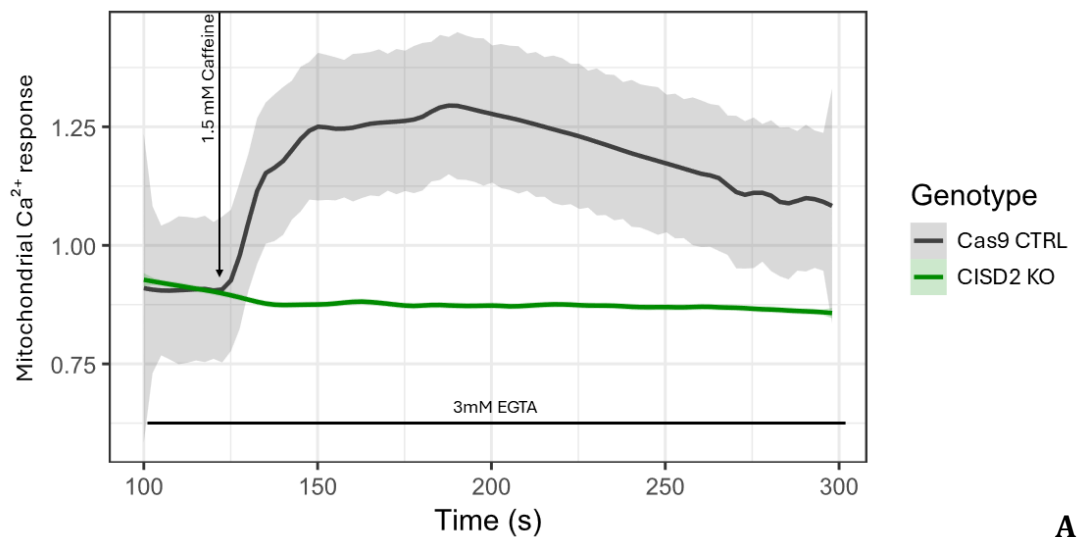
Regarding the impact of C1SD2^{FeSCD} and C1SD2^{N72S} on RyR3 activation, these mutants do not appear to differ significantly from C1SD2 WT [Fig 15, A; B]. Specifically, the amount of Ca^{2+} released in both mutant conditions is significantly different from the KO, but not from the WT, indicating that these mutations do not mimic the full effect of C1SD2 loss.

Furthermore, as previously done for measurements in C1SD2 KO and WT conditions, basal cytosolic Ca^{2+} levels were evaluated across all four conditions [Fig 15, C]. No significant differences were observed, consistent with previous results. Similarly, the time to peak was measured but showed no meaningful variation among the conditions [Fig 15, D].

In summary, these findings confirm that the absence of C1SD2 enhances RyR3-mediated Ca^{2+} release, and that physiological cellular responses can be restored by re-expressing C1SD2 WT. Conversely, the loss-of-function mutants C1SD2^{FeSCD} and C1SD2^{N72S} do not replicate the effects of complete C1SD2 absence. This suggests that disruption of the 2Fe-2S domain may influence RyR3 in a different way or affect an entirely separate pathway, which warrants further investigation.

5.6 Mitochondrial Ca^{2+} in presence and absence of C1SD2

Since alterations in mitochondrial Ca^{2+} are associated with metabolic activity, it is useful to evaluate mitochondrial Ca^{2+} levels. The study was made using Green mtCEPIA in order to detect and measure Ca^{2+} changes in the mitochondria, after RyR3 caffeine stimulation, through fluorescence microscopy.



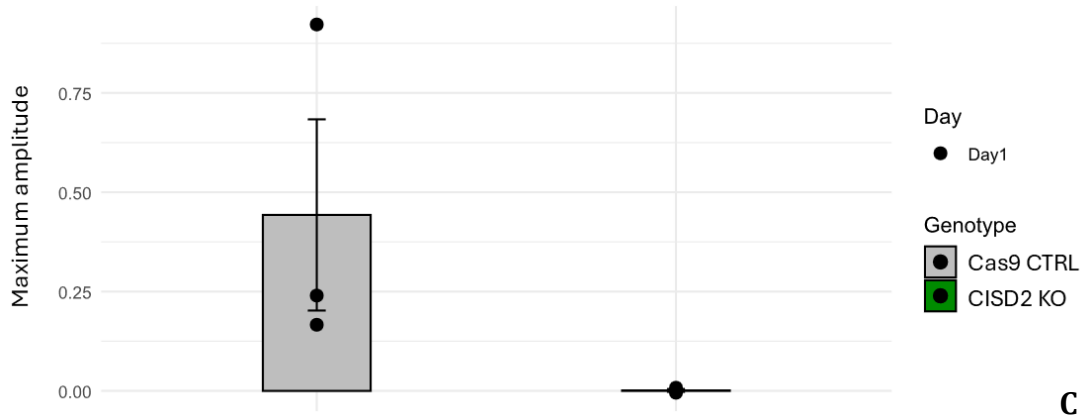


Figure 16. A, traces of mitochondrial Ca^{2+} dynamics in HEK293 RyR3 cells (grey, Cas9 CTRL) and HEK293 RyR3 C1SD2 KO cells (green, C1SD2 KO) following 1.5 mM caffeine stimulation of RyR3. Fluorescence was recorded using Green mtCEPIA (excitation 470 nm). Increased fluorescence corresponds to higher mitochondrial Ca^{2+} levels. **B-C**, Quantification of mitochondrial Ca^{2+} responses in HEK293 RyR3 cells (grey, Cas9 CTRL) and HEK293 RyR3 C1SD2 KO cells (green, C1SD2 KO) in the presence of 3 mM EGTA.

As shown in the traces in Figure 16A, following caffeine addition, mitochondrial Ca^{2+} levels increased only in the control condition with C1SD2 (grey, Cas9 CTRL), whereas a slight decrease was observed in the C1SD2 KO cells (green, C1SD2 KO), likely due to the presence of EGTA. This pattern is further confirmed by the AUC and maximum amplitude analyses, which show no detectable Ca^{2+} increase in the absence of C1SD2 (green, C1SD2 KO) (Fig. 16, B-C). It should be noted, as this experiment was performed only once, replication is necessary to obtain more robust and statistically reliable results.

Summarizing all the results, it can be concluded that C1SD2 acts as a negative regulator of RyR3 mediated Ca^{2+} release, reducing both the amplitude and kinetics of cytosolic Ca^{2+} elevation without affecting basal or ER Ca^{2+} levels. The absence of C1SD2 enhances RyR3 activity, while re-expression of the wild type protein restores normal Ca^{2+} dynamics. In contrast, the C1SD2^{FeSCD} and C1SD2^{N72S} mutants fail to reproduce the effects of C1SD2 loss, suggesting that these variants retain partial functionality or act through distinct mechanisms. In conclusion, these results indicate that, in the absence of C1SD2, Ca^{2+} does not enter mitochondria from the ER upon RyR3 activation. This effect may be related to a disruption of the MAMs complex, impairing normal Ca^{2+} uptake by the mitochondria.

6. Acknowledgements

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